CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: Validating an Interpretable Machine Learning Framework for Emergency Department Triage at Mercer General: A Caribbean Context

Student Name: Eliana Dookhoo

Statement of the Problem:
Emergency departments globally face persistent challenges of overcrowding, mistriage, and inefficient resource allocation, directly worsening patient outcomes and straining limited clinical resources *(Ariss et al., 2024; Chen et al., 2023; Xiao et al., 2023)*. While machine learning models have demonstrated strong predictive performance for triage disposition, ICU admission, wait times and length of stay *(Chang et al., 2024; Pandey et al., 2024; Song et al., 2025; Wang et al., 2025),* all were validated exclusively in Western, East Asian, and Australian contexts, leaving Caribbean and low-resource emergency settings unrepresented. This study proposes validating an interpretable ML triage framework at Mercer General, addressing generalisability and health equity gaps identified across the reviewed literature.

Previous Work:
Substantial research over the past decade has applied machine learning to emergency department operations with promising results. Ariss et al. (2024) developed 144 voting ensemble models using the MIMIC-IV database to predict the full spectrum of ED resource needs, achieving an average AUC of 0.82, demonstrating that data-driven tools can meaningfully support triage resource allocation(1). Chang et al. (2024) integrated structured triage data with NLP-processed free-text nursing notes to predict patient disposition in Taiwanese EDs, with a Random Forest model outperforming emergency physicians on F1 score, demonstrating that combining structured and unstructured data significantly improves predictive performance.(2)

For ICU-specific prediction, Chen et al. (2023) conducted a systematic review confirming that ML algorithms including Logistic Regression, Gradient Boosting, and Random Forest achieved AUC values ranging from 0.68 to 0.97 in identifying critically ill ED patients(3), while Pandey et al. (2024) developed an early ICU admission prediction model achieving an AUROC of 0.92, identifying triage category, ambulance arrival, and qSOFA score as the strongest predictors at the 30-minute mark(4). Complementing these findings, Lee et al. (2021) demonstrated that a neural network model trained on just eleven triage-available variables could predict hospitalization for urgent level-3 ED patients with a validation AUC of 0.80, highlighting the viability of parsimonious models built on readily accessible clinical data.(5)

Regarding operational metrics, Wang et al. (2025) demonstrated acceptable ML performance in predicting prolonged ED wait times, identifying ED crowding and mode of arrival as top predictors(6), while Song et al. (2025) developed interpretable models predicting both length of stay and disposition decision with accuracies exceeding 60% across ternary classifications(7). Building on the importance of rigorous data preprocessing, Naemi et al. (2021) quantified the impact of addressing missing values and data skewness in LOS prediction, demonstrating that applying multivariate imputation and resampling techniques improved ML model performance by an average of 15.75% in AUC and 32.19% in R^2 score, underscoring that data quality challenges must be explicitly addressed before deploying predictive models in clinical settings.(8) Xiao et al. (2023) further advanced the field by proposing TransNet and TextRNN models that reduced under-triage by 12.06% and over-triage by 17.92% compared to manual triage. (9)

Across all reviewed studies, however, models were built and validated exclusively in high-income, non-Caribbean settings, and none incorporated a fairness or health equity evaluation across demographic subgroups.

Proposed Solution:

This study proposes the development and validation of an interpretable machine learning framework tailored to the ED environment at Mercer General. Retrospective ED visit data will be extracted from Mercer General's electronic medical record system, incorporating structured triage variables; including vital signs, triage acuity, mode of arrival, demographics, and comorbidities, alongside unstructured free-text chief complaint notes processed through natural language processing.

Multiple ML algorithms will be trained and compared, including Random Forest, XGBoost, and Logistic Regression, with model interpretability ensured through SHAP values and feature importance analysis. The primary prediction targets will be clinical disposition, ICU admission likelihood, and prolonged wait time, evaluated using AUROC, F1 score, and Brier score.

Critically, this study will extend beyond aggregate performance metrics by conducting a subgroup fairness evaluation across race, ethnicity, language, and socioeconomic variables, directly addressing the health equity gap absent from all reviewed studies. The transparent framework will be designed for institutional adaptability, consistent with the approach recommended by *Song et al. (2025)*, enabling other Caribbean and low-resource hospitals to replicate the methodology using their own datasets.

References:

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